

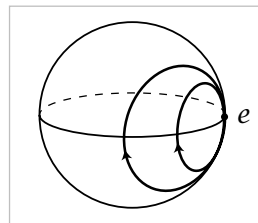
2. Using the Künneth formula, show that $\Sigma K(\mathbb{Z}_m \times \mathbb{Z}_n, 1) \simeq \Sigma K(\mathbb{Z}_m, 1) \vee \Sigma K(\mathbb{Z}_n, 1)$ if m and n are relatively prime. Thus to determine stable splittings of $K(\mathbb{Z}_n, 1)$ it suffices to do the case that n is a prime power, as in Proposition 4L3.
3. Extending Proposition 4L3, show that the $(2k+1)$ -skeleton of the suspension of a lens space with fundamental group of order p^n is homotopy equivalent to the wedge sum of the $(2k+1)$ -skeleta of the spaces X_i , if these X_i 's are chosen to have the minimum number of cells in each dimension, as described in the remarks following the proof.

4.J The Loopspace of a Suspension

Loopspaces appear at first glance to be hopelessly complicated objects, but if one is only interested in homotopy type, there are many cases when great simplifications are possible. One of the nicest of these cases is the loopspace of a sphere. We show in this section that ΩS^{n+1} has the weak homotopy type of the James reduced product $J(S^n)$ introduced in §3.2. More generally, we show that $\Omega \Sigma X$ has the weak homotopy type of $J(X)$ for every connected CW complex X . If one wants, one can strengthen 'weak homotopy type' to 'homotopy type' by quoting Milnor's theorem, mentioned in §4.3, that the loopspace of a CW complex has the homotopy type of a CW complex.

Part of the interest in $\Omega \Sigma X$ can be attributed to its close connection with the suspension homomorphism $\pi_i(X) \rightarrow \pi_{i+1}(\Sigma X)$. We will use the weak homotopy equivalence of $\Omega \Sigma X$ with $J(X)$ to give another proof that the suspension homomorphism is an isomorphism in dimensions up to approximately double the connectivity of X . In addition, we will obtain an exact sequence that measures the failure of the suspension map to be an isomorphism in dimensions between double and triple the connectivity of X . An easy application of this, together with results proved elsewhere in the book, will be to compute $\pi_{n+1}(S^n)$ and $\pi_{n+2}(S^n)$ for all n .

As a rough first approximation to ΩS^{n+1} there is a natural inclusion of S^n into ΩS^{n+1} obtained by regarding S^{n+1} as the reduced suspension ΣS^n , the quotient $(S^n \times I) / (S^n \times \partial I \cup \{e\} \times I)$ where e is the basepoint of S^n , then associating to each point $x \in S^n$ the loop $\lambda(x)$ in ΣS^n given by $t \mapsto (x, t)$. The figure shows what a few such loops look like. However, we cannot expect this inclusion $S^n \hookrightarrow \Omega S^{n+1}$ to be a homotopy equivalence since ΩS^{n+1} is an H-space but S^n is only an H-space when $n = 1, 3, 7$ by the theorem of Adams discussed in §4.B. The simplest way to correct this deficiency in S^n would be to replace it by the free H-space that it generates, the reduced product $J(S^n)$. Re-



call from §3.2 that a point in $J(S^n)$ is a formal product $x_1 \cdots x_k$ of points $x_i \in S^n$, with the basepoint e acting as an identity element for the multiplication obtained by juxtaposition of formal products. We would like to define a map $\lambda: J(S^n) \rightarrow \Omega S^{n+1}$ by setting $\lambda(x_1 \cdots x_k) = \lambda(x_1) \cdots \lambda(x_k)$, the product of the loops $\lambda(x_i)$. The only difficulty is in the parametrization of this product, which needs to be adjusted so that λ is continuous. The problem is that when some x_i approaches the basepoint $e \in S^n$, one wants the loop $\lambda(x_i)$ to disappear gradually from the product $\lambda(x_1) \cdots \lambda(x_k)$, without disrupting the parametrization as simply deleting $\lambda(e)$ would do. This can be achieved by first making the time it takes to traverse each loop $\lambda(x_i)$ equal to the distance from x_i to the basepoint of S^n , then normalizing the resulting product of loops so that it takes unit time, giving a map $I \rightarrow \Sigma S^n$.

More generally, this same procedure defines a map $\lambda: J(X) \rightarrow \Omega \Sigma X$ for any connected CW complex X , where ‘distance to the basepoint’ is replaced by any map $d: X \rightarrow [0, 1]$ with $d^{-1}(0) = e$, the basepoint of X .

Theorem 4J.1. *The map $\lambda: J(X) \rightarrow \Omega \Sigma X$ is a weak homotopy equivalence for every connected CW complex X .*

Proof: The main step will be to compute the homology of $\Omega \Sigma X$. After this is done, it will be easy to deduce that λ induces an isomorphism on homology using the calculation of the homology of $J(X)$ in Proposition 3C.8, and from this conclude that λ is a weak homotopy equivalence. It will turn out to be sufficient to consider homology with coefficients in a field F . We know that $H_*(J(X); F)$ is the tensor algebra $T\tilde{H}_*(X; F)$ by Proposition 3C.8, so we want to show that $H_*(\Omega \Sigma X; F)$ has this same structure, a result first proved in [Bott & Samelson 1953].

Let us write the reduced suspension $Y = \Sigma X$ as the union of two reduced cones $Y_+ = C_+ X$ and $Y_- = C_- X$ intersecting in the equatorial $X \subset \Sigma X$. Consider the path fibration $p: PY \rightarrow Y$ with fiber ΩY . Let $P_+ Y = p^{-1}(Y_+)$ and $P_- Y = p^{-1}(Y_-)$, so $P_+ Y$ consists of paths in Y starting at the basepoint and ending in Y_+ , and similarly for $P_- Y$. Then $P_+ Y \cap P_- Y$ is $p^{-1}(X)$, the paths from the basepoint to X . Since Y_+ and Y_- are deformation retracts of open neighborhoods U_+ and U_- in Y such that $U_+ \cap U_-$ deformation retracts onto $Y_+ \cap Y_- = X$, the homotopy lifting property implies that $P_+ Y$, $P_- Y$, and $P_+ Y \cap P_- Y$ are deformation retracts, in the weak sense, of open neighborhoods $p^{-1}(U_+)$, $p^{-1}(U_-)$, and $p^{-1}(U_+) \cap p^{-1}(U_-)$, respectively. Therefore there is a Mayer-Vietoris sequence in homology for the decomposition of PY as $P_+ Y \cup P_- Y$. Taking reduced homology and using the fact that PY is contractible, this gives an isomorphism

$$(i) \quad \Phi: \tilde{H}_*(P_+ Y \cap P_- Y; F) \xrightarrow{\cong} \tilde{H}_*(P_+ Y; F) \oplus \tilde{H}_*(P_- Y; F)$$

The two coordinates of Φ are induced by the inclusions, with a minus sign in one case, but Φ will still be an isomorphism if this minus sign is eliminated, so we may assume this has been done.

We claim that the isomorphism Φ can be rewritten as an isomorphism

$$(ii) \quad \Theta: \tilde{H}_*(\Omega Y \times X; F) \xrightarrow{\sim} \tilde{H}_*(\Omega Y; F) \oplus \tilde{H}_*(\Omega Y; F)$$

To see this, we observe that the fibration $P_+Y \rightarrow Y_+$ is fiber-homotopically trivial. This is true since the cone Y_+ is contractible, but we shall need an explicit fiber homotopy equivalence $P_+Y \simeq \Omega Y \times Y_+$, and this is easily constructed as follows. Define $f_+: P_+Y \rightarrow \Omega Y \times Y_+$ by $f_+(y) = (y \cdot y_y^+, y)$ where $y = y(1)$ and y_y^+ is the obvious path in Y_+ from $y = (x, t)$ to the basepoint along the segment $\{x\} \times I$. In the other direction, define $g_+: \Omega Y \times Y_+ \rightarrow P_+Y$ by $g_+(y, y) = y \cdot \bar{y}_y^+$ where the bar denotes the inverse path. Then f_+g_+ and g_+f_+ are fiber-homotopic to the respective identity maps since $\bar{y}_y^+ \cdot y_y^+$ and $y_y^+ \cdot \bar{y}_y^+$ are homotopic to the constant paths.

In similar fashion the fibration $P_-Y \rightarrow Y_-$ is fiber-homotopically trivial via maps f_- and g_- . By restricting a fiber-homotopy trivialization of either P_+Y or P_-Y to $P_+Y \cap P_-Y$, we see that the fibration $P_+Y \cap P_-Y$ is fiber-homotopy equivalent to the product $\Omega Y \times X$. Let us do this using the fiber-homotopy trivialization of P_-Y . The groups in (i) can now be replaced by those in (ii). The map Φ has coordinates induced by inclusion, and it follows that the corresponding map Θ in (ii) has coordinates induced by the two maps $\Omega Y \times X \rightarrow \Omega Y$, $(y, x) \mapsto y \cdot \lambda(x)$ and $(y, x) \mapsto y$. Namely, the first coordinate of Θ is induced by $f_+g_-|_{\Omega Y \times X}$ followed by projection to ΩY , and the second coordinate is the same but with f_-g_- in place of f_+g_- .

Writing the two coordinates of Θ as Θ_1 and Θ_2 , the fact that Θ is an isomorphism means that the restriction of Θ_1 to the kernel of Θ_2 is an isomorphism. Via the Künneth formula we can write $\tilde{H}_*(\Omega Y \times X; F)$ as $(H_*(\Omega Y; F) \otimes \tilde{H}_*(X; F)) \oplus \tilde{H}_*(\Omega Y; F)$ where projection onto the latter summand is Θ_2 . Hence Θ_1 gives an isomorphism from the first summand $H_*(\Omega Y; F) \otimes \tilde{H}_*(X; F)$ onto $\tilde{H}_*(\Omega Y; F)$. Since $\Theta_1(y, x) = (y \cdot \lambda(x))$, this means that the composed map

$$H_*(\Omega Y; F) \otimes \tilde{H}_*(X; F) \xrightarrow{\mathbb{1} \otimes \lambda_*} H_*(\Omega Y; F) \otimes \tilde{H}_*(\Omega Y; F) \longrightarrow \tilde{H}_*(\Omega Y; F)$$

with the second map Pontryagin product, is an isomorphism. Now to finish the calculation of $H_*(\Omega Y; F)$ as the tensor algebra $T\tilde{H}_*(X; F)$, we apply the following algebraic lemma, with $A = H_*(\Omega Y; F)$, $V = \tilde{H}_*(X; F)$, and $i = \lambda_*$.

Lemma 4J.2. *Let A be a graded algebra over a field F with $A_0 = F$ and let V be a graded vector space over F with $V_0 = 0$. Suppose we have a linear map $i: V \rightarrow A$ preserving grading, such that the multiplication map $\mu: A \otimes V \rightarrow \tilde{A}$, $\mu(a \otimes v) = ai(v)$, is an isomorphism. Then the canonical algebra homomorphism $i: TV \rightarrow A$ extending the previous i is an isomorphism.*

For example, if V is a 1-dimensional vector space over F , as happens in the case $X = S^n$, then this says that if the map $A \rightarrow \tilde{A}$ given by right-multiplication by an element $a = i(v)$ is an isomorphism, then A is the polynomial algebra $F[a]$. The

general case can be viewed as the natural generalization of this to polynomials in any number of noncommuting variables.

Proof: Since μ is an isomorphism, each element $a \in A_n$ with $n > 0$ can be written uniquely in the form $\mu(\sum_j a_j \otimes v_j) = \sum_j a_j i(v_j)$ for $v_j \in V$ and $a_j \in A_{n(j)}$, with $n(j) < n$ since $V_0 = 0$. By induction on n , $a_j = i(\alpha_j)$ for a unique $\alpha_j \in (TV)_{n(j)}$. Thus $a = i(\sum_j \alpha_j \otimes v_j)$ so i is surjective. Since these representations are unique, i is also injective. The induction starts with the hypothesis that $A_0 = F$, the scalars in TV . $\square$

Returning now to the proof of the theorem, we observe that λ is an H-map: The two maps $J(X) \times J(X) \rightarrow \Omega\Sigma X$, $(x, y) \mapsto \lambda(xy)$ and $(x, y) \mapsto \lambda(x)\lambda(y)$, are homotopic since the loops $\lambda(xy)$ and $\lambda(x)\lambda(y)$ differ only in their parametrizations. Since λ is an H-map, the maps $X \hookrightarrow J(X) \xrightarrow{\lambda} \Omega\Sigma X$ induce the commutative diagram at the right. We have shown that the downward map on the right is an isomorphism, and the same is true of the one on the left by the calculation of $H_*(J(X); F)$ in Proposition 3C.8. Hence λ_* is an isomorphism. By Corollary 3A.7 this is also true for $\mathbb{Z}$ coefficients. When X is simply-connected, so are $J(X)$ and $\Omega\Sigma X$, so after taking a CW approximation to $\Omega\Sigma X$, Whitehead's theorem implies that λ is a weak homotopy equivalence. In the general case that X is only connected, we obtain the same conclusion by applying the generalization of Whitehead's theorem to abelian spaces, Proposition 4.74, since $J(X)$ and $\Omega\Sigma X$ are H-spaces, hence have trivial action of π_1 on all homotopy groups. $\square$

$$\begin{array}{ccc} & T\tilde{H}_*(X; F) & \\ \swarrow & & \searrow \\ H_*(J(X); F) & \xrightarrow{\lambda_*} & H_*(\Omega\Sigma X; F) \end{array}$$

Using the natural identification $\pi_i(\Omega\Sigma X) = \pi_{i+1}(\Sigma X)$, the inclusion $X \hookrightarrow \Omega\Sigma X$ induces the suspension map $\pi_i(X) \rightarrow \pi_{i+1}(\Sigma X)$. Since this inclusion factors through $J(X)$, we can identify the relative groups $\pi_i(\Omega\Sigma X, X)$ with $\pi_i(J(X), X)$. If X is n -connected then the pair $(J(X), X)$ is $(2n + 1)$ -connected since we can replace X by a complex with n -skeleton a point, and then the $(2n + 1)$ -skeleton of $J(X)$ is contained in X . Thus we have:

Corollary 4J.3. *The suspension map $\pi_i(X) \rightarrow \pi_{i+1}(\Sigma X)$ for an n -connected CW complex X is an isomorphism if $i \leq 2n$ and a surjection if $i = 2n + 1$. $\square$*

In the case of a sphere we can describe what happens in the first dimension when suspension is not an isomorphism, namely the suspension $\pi_{2n-1}(S^n) \rightarrow \pi_{2n}(S^{n+1})$ which the corollary guarantees only to be a surjection. The CW structure on $J(S^n)$ consists of a single cell in each dimension a multiple of n , so from exactness of $\pi_{2n}(J(S^n), S^n) \xrightarrow{\partial} \pi_{2n-1}(S^n) \xrightarrow{\Sigma} \pi_{2n}(S^{n+1})$ we see that the kernel of the suspension $\pi_{2n-1}(S^n) \rightarrow \pi_{2n}(S^{n+1})$ is generated by the attaching map of the $2n$ -cell of $J(S^n)$. This attaching map is the Whitehead product $[\mathbb{1}, \mathbb{1}]$, as we noted in §4.2 when we

defined Whitehead products following Example 4.52. When n is even, the Hopf invariant homomorphism $\pi_{2n-1}(S^n) \rightarrow \mathbb{Z}$ has the value ± 2 on $[\mathbb{1}, \mathbb{1}]$, as we saw in §4.B. If there is no map of Hopf invariant ± 1 , it follows that $[\mathbb{1}, \mathbb{1}]$ generates a $\mathbb{Z}$ summand of $\pi_{2n-1}(S^n)$, and so the suspension homomorphism simply cancels this summand from $\pi_{2n-1}(S^n)$. By Adams' theorem, this is the situation for all even n except 2, 4, and 8.

When $n = 2$ we have $\pi_3(S^2) \approx \mathbb{Z}$ generated by the Hopf map η with Hopf invariant 1, so $2\eta = \pm[\mathbb{1}, \mathbb{1}]$, generating the kernel of the suspension, hence:

Corollary 4J.4. $\pi_{n+1}(S^n)$ is $\mathbb{Z}_2$ for $n \geq 3$, generated by the suspension or iterated suspension of the Hopf map. $\square$

The situation for $n = 4$ and 8 is more subtle. We do not have the tools available here to do the actual calculation, but if we consult the table near the beginning of §4.1 we see that the suspension $\pi_7(S^4) \rightarrow \pi_8(S^5)$ is a map $\mathbb{Z} \oplus \mathbb{Z}_{12} \rightarrow \mathbb{Z}_{24}$. By our preceding remarks we know this map is surjective with kernel generated by the single element $[\mathbb{1}, \mathbb{1}]$. Algebraically, what must be happening is that the coordinate of $[\mathbb{1}, \mathbb{1}]$ in the $\mathbb{Z}$ summand is twice a generator, while the coordinate in the $\mathbb{Z}_{12}$ summand is a generator. Thus a generator of the $\mathbb{Z}$ summand, which we may take to be the Hopf map $S^7 \rightarrow S^4$, suspends to a generator of the $\mathbb{Z}_{24}$. For $n = 8$ the situation is entirely similar, with the suspension $\pi_{15}(S^8) \rightarrow \pi_{16}(S^9)$ a homomorphism $\mathbb{Z} \oplus \mathbb{Z}_{120} \rightarrow \mathbb{Z}_{240}$.

We can also obtain some information about suspension somewhat beyond the edge of the stable dimension range. Since S^n is $(n-1)$ -connected and $(J(S^n), S^n)$ is $(2n-1)$ -connected, we have isomorphisms $\pi_i(J(S^n), S^n) \approx \pi_i(J(S^n)/S^n)$ for $i \leq 3n-2$ by Proposition 4.28. The group $\pi_i(J(S^n)/S^n)$ is isomorphic to $\pi_i(S^{2n})$ in the same range $i \leq 3n-2$ since $J(S^n)/S^n$ has S^{2n} as its $(3n-1)$ -skeleton. Thus the terminal portion of the long exact sequence of the pair $(J(S^n), S^n)$ starting with the term $\pi_{3n-2}(S^n)$ can be written in the form

$$\pi_{3n-2}(S^n) \xrightarrow{\Sigma} \pi_{3n-1}(S^{n+1}) \rightarrow \pi_{3n-2}(S^{2n}) \rightarrow \pi_{3n-3}(S^n) \xrightarrow{\Sigma} \pi_{3n-2}(S^{n+1}) \rightarrow \dots$$

This is known as the **EHP sequence** since its three maps were originally called E , H , and P . (The German word for ‘suspension’ begins with E, the H refers to a generalization of the Hopf invariant, and the P denotes a connection with Whitehead products; see [Whitehead 1978] for more details.) Note that the terms $\pi_i(S^{2n})$ in the EHP sequence are stable homotopy groups since $i \leq 3n-2$. Thus we have the curious situation that stable homotopy groups are measuring the lack of stability of the groups $\pi_i(S^n)$ in the range $2n-1 \leq i \leq 3n-2$, the so-called *metastable* range.

Specializing to the first interesting case $n = 2$, the sequence becomes

$$\begin{array}{ccccccc} \pi_4(S^2) & \xrightarrow{\Sigma} & \pi_5(S^3) & \longrightarrow & \pi_4(S^4) & \longrightarrow & \pi_3(S^2) \xrightarrow{\Sigma} \pi_4(S^3) \longrightarrow 0 \\ \parallel & & & & \parallel & & \parallel \\ \mathbb{Z}_2 & & & & \mathbb{Z} & & \mathbb{Z}_2 \end{array}$$

From the Hopf bundle $S^1 \rightarrow S^3 \rightarrow S^2$ we have $\pi_4(S^2) \approx \pi_4(S^3) \approx \mathbb{Z}_2$, with $\pi_4(S^2)$ generated by the composition $\eta(\Sigma\eta)$ where η is the Hopf map $S^3 \rightarrow S^2$. From exactness of the latter part of the sequence we deduce that the map $\pi_4(S^4) \rightarrow \pi_3(S^2)$ is injective, and hence that the suspension $\pi_4(S^2) \rightarrow \pi_5(S^3)$ is surjective, so $\pi_5(S^3)$ is either $\mathbb{Z}_2$ or 0. From the general suspension theorem, the suspension $\pi_5(S^3) \rightarrow \pi_6(S^4)$ is surjective as well, and the latter group is in the stable range. We show in Proposition 4L.11 that the stable group π_2^S is nonzero, and so we conclude that $\pi_{n+2}(S^n) \approx \mathbb{Z}_2$ for all $n \geq 2$, generated by the composition $(\Sigma^{n-2}\eta)(\Sigma^{n-1}\eta)$.

We will see in [SSAT] that the EHP sequence extends all the way to the left to form an infinite exact sequence when n is odd, and when n is even a weaker statement holds: The sequence extends after factoring out all odd torsion.

Replacing S^n by any $(n-1)$ -connected CW complex X , our derivation of the finite EHP sequence generalizes immediately to give an exact sequence

$$\pi_{3n-2}(X) \xrightarrow{\Sigma} \pi_{3n-1}(\Sigma X) \rightarrow \pi_{3n-2}(X \wedge X) \rightarrow \pi_{3n-3}(X) \xrightarrow{\Sigma} \pi_{3n-2}(\Sigma X) \rightarrow \cdots$$

using the fact that $J_2(X)/X = X \wedge X$.

The generalization of the results of this section to $\Omega^n \Sigma^n X$ turns out to be of some importance in homotopy theory. In case we do not get to this topic in [SSAT], the reader can begin to learn about it by looking at [Carlsson & Milgram 1995].

Exercise

1. Show that $\Omega \Sigma X$ for a nonconnected CW complex X reduces to the connected case by showing that each path-component of $\Omega \Sigma X$ is homotopy equivalent to $\Omega \Sigma(\bigvee_{\alpha} X_{\alpha})$ where the X_{α} 's are the components of X .

4.K The Dold–Thom Theorem

In the preceding section we studied the free monoid $J(X)$ generated by a space X , and in this section we take up its commutative analog, the free abelian monoid generated by X . This is the infinite symmetric product $SP(X)$ introduced briefly in §3.C. The main result will be a theorem of [Dold & Thom 1958] asserting that $\pi_* SP(X) \approx \tilde{H}_*(X; \mathbb{Z})$ for all connected CW complexes X . In particular this yields the surprising fact that $SP(S^n)$ is a $K(\mathbb{Z}, n)$, and more generally that the functor SP takes Moore spaces $M(G, n)$ to Eilenberg–MacLane spaces $K(G, n)$. This leads to the general result that for all connected CW complexes X , $SP(X)$ has the homotopy type of a product of Eilenberg–MacLane spaces. In other words, the k -invariants of $SP(X)$ are all trivial.